PRODUCTION INFRASTRUCTURE AUDIT REPORT

Proxmox Cluster, Ceph Storage and ZFS RAIDZ2 Validation

Document owner	Nishu Saini
Function	CISAI IT Support / Infrastructure
Environment	cisai-pve three-node cluster
Nodes	prox22, prox23, prox24
Validation date	05 September 2026
Classification	Internal - Production Audit Evidence
Document version	1.0

OVERALL CONTROL RESULT: PASS

No infrastructure failure is evidenced in the supplied final validation outputs. Two Proxmox warnings identify tests not executed in read-only mode.

Prepared from system-generated validation logs supplied by the document owner. This report summarizes recorded evidence and does not replace the underlying logs.

1. Executive Summary

The supplied evidence records a healthy three-node Proxmox cluster, healthy Ceph services and three ONLINE node-local ZFS RAIDZ2 pools. The Proxmox validation on prox24 recorded 42 passed checks, 2 warnings and 0 failures. The warnings state that live migration and disruptive HA failover simulation were not executed in read-only mode.

Control area	Recorded result	Audit disposition
Cluster quorum	3 nodes; Quorate: Yes	PASS
Ceph health	HEALTH_OK; 12 OSDs up/in; 33 PGs active+clean	PASS
ZFS RAIDZ2	Three node-local pools ONLINE; 7 members each	PASS
ZFS error counters	READ 0, WRITE 0, CKSUM 0; no known data errors	PASS
Integrity testing	1 GiB test files; SHA-256 readback verified	PASS
Scrub testing	Completed with 0 errors and repaired 0B	PASS
VM live migration	Not executed in cited read-only Proxmox run	OPEN TEST
Disruptive HA failover	Not executed in cited read-only Proxmox run	OPEN TEST

Audit conclusion

Based strictly on the supplied outputs, the tested cluster, Ceph and ZFS storage controls passed the recorded health and integrity checks. Production readiness sign-off for VM mobility and failover remains subject to separate, controlled execution evidence because those actions were explicitly skipped in the read-only Proxmox validation.

2. Scope and Evidence Basis

Evidence item	Coverage
Proxmox validation log	Cluster, quorum, services, Ceph, ZFS visibility, shared storage and HA configuration
zfs_raidz2_combined_prox22.log	prox22 RAIDZ2 health, properties, SMART, I/O, integrity and scrub evidence
zfs_raidz2_combined_prox23.log	prox23 RAIDZ2 health, properties, SMART, I/O, integrity and scrub evidence
zfs_raidz2_combined_prox24.log	prox24 RAIDZ2 health, properties, SMART, I/O, integrity and scrub evidence

Scope limitation: The PDF is a summarized audit report. Full command output, device WWNs, event streams and checksums remain in the source logs. No independent re-execution of commands was performed for this report.

3. Proxmox Cluster and Ceph Validation

Item	Recorded evidence	Result
Cluster	cisai-pve, config version 3, knet transport	PASS
Membership	prox24, prox22 and prox23; total votes 3	PASS
Quorum	Expected votes 3; quorum 2; Quorate: Yes	PASS
Core services	pve-cluster active; corosync active; no recent Corosync warning/error entries	PASS
Platform	Proxmox VE 9.2.0; pve-manager 9.2.11; kernel 7.0.14-14-pve	RECORDED
Time	Asia/Kolkata; NTP synchronized	PASS

Ceph control	Recorded evidence	Result
Overall health	HEALTH_OK	PASS
MON	3 daemons in quorum: prox24, prox22, prox23	PASS
MGR	prox23 active; prox24 and prox22 standby	PASS
OSD	12 total, 12 up, 12 in	PASS
Placement groups	33 active+clean	PASS
Raw capacity	218 TiB total; 588 MiB used in cited run	RECORDED
OSD latency indicator	Commit/apply values shown as 0 ms in cited run	PASS
RBD storage	ceph storage listed active by pvesm	PASS

4. ZFS RAIDZ2 Validation

Node / pool	Layout and capacity	Integrity status	Result
prox22 / local-zfs-prox22	RAIDZ2, 7 x 18.2T members, 127T pool, 0% fragmentation	ONLINE; 0/0/0 errors; no known data errors	PASS
prox23 / local-zfs-prox23	RAIDZ2, 7 x 18.2T members, 127T pool, 0% fragmentation	ONLINE; 0/0/0 errors; no known data errors	PASS
prox24 / local-zfs-prox24	RAIDZ2, 7 x 18.2T members, 127T pool, 0% fragmentation	ONLINE; 0/0/0 errors; no known data errors	PASS

Common recorded properties

The logs record ashift=12, lz4 compression, checksum enabled, atime off, sync=standard, xattr=sa, and zfs-zed active. Each pool includes a node-local pve dataset with mountpoint=none, consistent with its use as Proxmox ZFS storage.

5. Controlled Data Integrity and Scrub Evidence

Node	Controlled I/O evidence	Direct read	Scrub evidence	Result
prox22	1 GiB test; SHA-256 readback OK	1.1 GB/s recorded	Repaired 0B; 0 errors	PASS
prox23	1 GiB test; SHA-256 readback OK	1.1 GB/s recorded	Repaired 0B; 0 errors	PASS
prox24	1 GiB test; SHA-256 readback OK	1.0-1.1 GB/s recorded	Repaired 0B; 0 errors	PASS

Performance interpretation note

The displayed throughput values are point-in-time results from controlled 1 GiB file operations in the supplied logs. They are evidence of successful test completion, not a capacity guarantee, service-level commitment or statistically controlled performance baseline.

6. Exceptions, Warnings and Open Items

ID	Observation	Risk interpretation	Required evidence / action
OBS-01	VM live migration was not executed in the read-only validation.	VM mobility was not demonstrated by this report.	Execute a controlled migration of an approved test VM and retain source/target, start/end status and guest connectivity evidence.
OBS-02	Disruptive node-failure simulation was not performed.	Automatic recovery under node loss was not demonstrated by this report.	Execute only under an approved maintenance plan with fencing, quorum, console and rollback controls; retain HA recovery evidence.
INFO-01	Other nodes local ZFS stores appear disabled on prox24.	Expected for node-restricted local storage; no defect evidenced.	No corrective action required if /etc/pve/storage.cfg node restrictions match design.

7. Audit Sign-off

The following sign-off records review and acceptance of the evidence and of the stated scope limitations.

Role	Name	Signature	Date
Prepared by	Nishu Saini		
Technical reviewer			
Change owner / approver			

AUDIT DISPOSITION: PASS WITH OPEN RESILIENCE TESTS

Cluster, Ceph and ZFS health/integrity controls passed in the supplied evidence. VM live migration and disruptive HA failover require separate controlled evidence for full resilience closure.

Appendix A - Evidence Traceability

Claim	Source evidence
prox22 pool ONLINE, RAIDZ2, seven members, zero errors	zfs_raidz2_combined_prox22.log: pool configuration and health sections
prox23 pool ONLINE, RAIDZ2, seven members, zero errors	zfs_raidz2_combined_prox23.log: pool configuration and health sections
prox24 pool ONLINE, RAIDZ2, seven members, zero errors	zfs_raidz2_combined_prox24.log: pool configuration and health sections
Integrity readback and scrub success	Controlled data integrity, SHA-256 and scrub sections in each combined node log
Cluster and Ceph result	proxmox_validation_prox24_20260905_093952.log supplied in conversation
Open migration and failover tests	Warnings in Proxmox final summary

Appendix B - Retention Guidance

Retain this PDF together with the original logs and their SHA-256 checksum files in the approved audit repository. Preserve original filenames and timestamps. Any rerun should be stored as a new evidence set rather than overwriting prior evidence.